Eric J. Pabon Cancel

Curriculum Vitae

Contact Information

Mathematical Sciences Building, Office 609
Department of Mathematics, Purdue University

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Research Interests

Dynamical Systems (e.g. Mathematical Physics, Symplectic Geometry, Modeling in Material Science), Machine Learning (e.g. Physics-Informed Neural Networks), Computational & Applied Mathematics (e.g. Numerical Methods, Mathematical Algorithms, Data Assimilation).

Education & Background

Doctor of Philosophy in Mathematics May 2029
Purdue University, West Lafayette, Indiana | Advisor: Prof. Aaron Yip

Master of Science in Mathematics May 2026
Purdue University, West Lafayette, Indiana

Bachelor of Science in Pure Mathematics (*Magna Cum Laude*) December 2022
Curricular Sequence in Applied Mathematics for Science and Engineering
University of Puerto Rico, Mayagüez Campus (UPRM), Mayagüez, Puerto Rico | Mentor: Reyes M. Ortiz Albino

Skills and Other Information

Programming & Computation: Python, Julia, C++, SageMath, MATLAB | **Formatting & Tools:** HTML, LaTeX, Git
Math & AI/ML: NumPy, SciPy, PyTorch, TensorFlow, JAX, Flux, CUDA | **Spoken Languages:** English and Spanish

Research Experience

URA-Sandia Graduate Student Summer Fellow
URA-Sandia Graduate Student Summer Fellowship (administered by ORISE and ORAU) May 2025–August 2025
Computational & Information Sciences Foundation, Sandia National Laboratories Livermore, California
Supervised by: Dr. Moe Khalil & Dr. Rileigh Bandy, Sandia National Laboratories
Research Topics: Machine Learning-based Surrogate Models, Data Assimilation

GEM Fellow Summer Research Program Intern
MIT Lincoln Laboratory Summer Research Program (GEM Fellowship Employer Sponsor) May 2023–August 2023
Group 39, Division 3, MIT Lincoln Laboratory, Massachusetts Institute of Technology Lexington, Massachusetts
Supervised by: Dr. Sam Polk & Dr. Mabel Ramírez, MIT Lincoln Laboratory
Research Topics: Classical Machine Learning, Bayesian Inference, Generative AI, Unsupervised Learning

Research Opportunity for Undergraduate Students in STEM
Puerto Rico Louis Stokes Alliance for Minority Participation August 2019–December 2022
Department of Mathematical Sciences, University of Puerto Rico, Mayagüez Campus Mayagüez, Puerto Rico
Supervised by: Prof. Reyes M. Ortiz Albino, University of Puerto Rico at Mayagüez
Research Topics: Number Theory, Commutative Algebra, Factorization Theory in Integral Domains

NSF REU Participant

- Summer@ICERM 2022: Computational Combinatorics June 2022–August 2022
  Institute for Computational and Experimental Research in Mathematics, Brown University Providence, Rhode Island
  Supervised by: Prof. Pamela E. Harris, University of Wisconsin-Milwaukee
  Research Topics: Enumerative Combinatorics, Parking Functions
- NSF REU in Combinatorics, Probability and Algebraic Coding Theory June 2021–August 2021
  East Tennessee State University & University of Puerto Rico at Ponce Johnson City, Tennessee
  Supervised by: Prof. Fernando Piñero González, University of Puerto Rico at Ponce
  Research Topics: Algebraic Geometry & Linear Error Correcting Codes

Projects

Project in Numerical Partial Differential Equations
Purdue University, West Lafayette January 2026–May 2026
MA61500: Numerical Methods for Partial Differential Equations | Instructor: Prof. Di Qi
TBD

Project in Numerical Differential Equations
Purdue University, West Lafayette — August 2025–December 2025
MA57300: Numerical Solutions of Ordinary Differential Equations | Instructor: Prof. Di Qi
Data Assimilation for the Lorenz 96 Model

Projects in Optimal Transport and Neural Networks
Purdue University, West Lafayette — January 2025–May 2025
MA59500OT: Computational Optimal Transport and Deep Generative Models | Instructor: Prof. Rongjie Lai
Normalizing Flows Optimal Transport implementation on MNIST Dataset
WGAN and Monge Map implementation on MNIST Dataset

Project in Neural Networks and Dynamical Systems
Purdue University, West Lafayette — November 2024–December 2024
MA59500MM: Introduction to Mathematical Modeling | Instructor: Prof. Alexandria Volkening
Physics-Informed Neural Networks (PINNs) for Hurricane Trajectory Prediction

Project in Biotechnology
MIT Lincoln Laboratory Summer Research Program — June 2023–July 2023
2023 MIT Lincoln Laboratory Intern Innovative Idea Challenge (I^3C) | Mentors: Ryan Burrow and Ashok Kumar
SKINS: Skin-growth boosting and Intra-absorptive Solution bandages (3rd Place Project Winner)

Awards

Fellowships, Scholarships and Prizes

2025 Universities Research Association-Sandia National Labs Graduate Summer Fellowship	May 2025–August 2025
2023 National GEM Consortium PhD Science Fellowship	August 2023–May 2024
• Purdue University Department of Mathematics Sponsorship	August 2023–May 2024
• MIT Lincoln Laboratory Employer Sponsorship (Internship)	May 2023–August 2023
2023 MIT Lincoln Laboratory I^3C 3^{rd} Place Research Proposal Prize	July 2023
2022 Evertec Inc. STEM Scholarship	October 2022
Puerto Rico-Louis Stokes Alliance for Minority Participation Research Scholarship	August 2019–December 2022

Merits and Honors

2026 URA STEM Ambassador	March 2026
2023 Ford Foundation Predoctoral Fellowship Honorable Mention	March 2023
2022 Hispanic Scholarship Fund Scholar	June 2022
National Math Alliance Predoctoral Scholar	November 2021
UPRM Faculty of Arts and Sciences Honor Roll	August 2018–May 2023
Puerto Rico Department of Education Math Olympiad, 2nd Place Regional Winner	February 2018
National Trig-Star Math Competition, 16th Overall Finalist	June 2017
Eagle Scout Rank, with 2 Silver Palms	May 2017
Puerto Rico Department of Education Math Olympiad, 1st Place District Winner	February 2016, 2017

Papers and Articles (The asterisk symbol (*) denotes alphabetical order authorship.)

Research Articles and Preprints:

[1] S. Polk, E.J. Pabon-Cancel, R. Paleja, K. Chestnut-Chang, R. Jensen and M. Ramirez.
Unsupervised Behavior Inference from Human Action Sequences (UNBIAS).
2024 IEEE Conference on Games (CoG), Milan, Italy, 2024, pp. 1-8.
DOI: https://doi.org/10.1109/CoG60054.2024.10645587

[2] *A. Allen, E.J. Pabon-Cancel, F. Piñero-Gonzalez and L. Polanco.
Improving the Dimension Bound of Hermitian-Lifted Codes.
arXiv: https://arxiv.org/abs/2302.01557

[3] *D. Chen, P.E. Harris, J. Carlos Martinez Mori, E.J. Pabon-Cancel and G. Sargent.
Permutation Invariant Parking Assortments.
Enumerative Combinatorics and Applications, **4:1**, 1-25 (2024). #S2R4.
DOI: https://doi.org/10.54550/ECA2024V4S1R4

[4] *I. Byrne, N. Dodson, R. Lynch, E.J. Pabon-Cancel and F. Piñero-Gonzalez.
Improving the minimum distance bound of Trace Goppa codes.

Designs, Codes and Cryptography. **91**, 2649–2663 (2023).
DOI: https://doi.org/10.1007/s10623-023-01216-6

Contributions to the profession:

[1] *P.E. Harris, Z. Markman, L. Martinez, A. Mock, E.J. Pabón-Cancel, A. Verga, and S. Wang.
A Model for a One-Hour Workshop on Mentoring.
MAA Focus, **43**(1), 18-21 (2023).

Service and Outreach

Academic Service

- Universities Research Association
  URA STEM Ambassadors Program — March 2026–present
  University Ambassador (Purdue University) — West Lafayette, Indiana

Teaching and Grading

- MA 59500MB: Mathematical Biology (Grading) — January 2026–May 2026
- MA 32500: History of Mathematics (Grading) — January 2026–May 2026
- MA 26100 REC: Multivariate Calculus Recitation (Teaching) — August 2025–December 2025
  January 2025–May 2025
  August 2024–December 2024
- MA 13900: Mathematics for Elementary Teachers III (Grading) — June 2024–August 2024

Student Associations

- Purdue SACNAS Chapter
  Purdue University — August 2025–May 2026
  Member — West Lafayette, Indiana
- PythagoRUM
  University of Puerto Rico, Mayagüez Campus — August 2022–December 2022
  Co-founder & Vice-President — Mayagüez, Puerto Rico
- Society of Physics Students, UPRM Chapter
  University of Puerto Rico, Mayagüez Campus — August 2018–December 2022
  Committee Assistant — Mayagüez, Puerto Rico

Poster Sessions, Panels, Presentations and Conferences

- Purdue University Student Computational and Applied Mathematics Seminar — 27 March 2026
  Helen B. Schleman Hall, Purdue University — West Lafayette, Indiana
  Panel: Internships and Job Oppoortunities Beyond Academia
- Purdue Graduate Student Government Research-O-Rama — 21 February 2026
  West Lafayette Public Library — West Lafayette, Indiana
  Poster: Data-Driven Closure Models (DDCMs)
- Purdue University Prospective Student Visit Weekend Graduate Student Research Expo. — 20 February 2026
  Mathematical Sciences Building, Purdue University — West Lafayette, Indiana
  Poster: Data-Driven Closure Models (DDCMs)
- Purdue University Student Computational and Applied Mathematics Seminar — 13 February 2026
  Helen B. Schleman Hall, Purdue University — West Lafayette, Indiana
  Presentation: Data-Driven Closure Models (DDCMs)
- URA-Sandia Graduate Summer Fellowship Lightning Talk — 6 August 2025
  Presentation: Data-Driven Closure Models (DDCMs) — Virtual Seminar
- Sandia National Laboratories CA SIP Intern Symposium — 5 August 2025
  Auditorium, Sandia National Laboratories-Livermore — Livermore, California
  Poster: Data-Driven Closure Models (DDCMs)
- Combinatorics and Coding Theory in the Tropics (UPR-Ponce) — 18 July 2025
  Invited REU Seminar: My Story & Permutation-Invariant Parking Assortments — Virtual Seminar

- Purdue University Student Commutative Algebra Seminar — 18 November 2024
  Helen B. Schleman Hall, Purdue University — West Lafayette, Indiana
  Presentation: Results in $\tau_{(n)}$-factorizations and $\tau_{(n)}$-primes.
- Purdue University Student Math History Seminar — 9 September 2024
  Lawson Computer Science Building, Purdue University — West Lafayette, Indiana
  Presentation: Testimonios: Stories of Latinos and Hispanics in Mathematics
- Underrepresented Students in Topology and Algebra Research Symposium 2024 — 20-21 April 2024
  University of Iowa — Iowa City, Iowa
- 2023 MIT Lincoln Lab Intern Innovative Idea Challenge — 14, 21 July 2023
  MIT Lincoln Laboratory Auditorium — Lexington, Massachusetts
  Poster: Skin-Absorptive and Skin-Growth Boosting Bandages
  Presentation: SKINS: Skin-growth boosting and Intra-absorptive Solution Bandages
- Combinatorics and Coding Theory in the Tropics (UPR-Ponce) — 7 July 2023
  Invited REU Seminar Talk: Graduate School: Application tips and advice — Virtual Seminar
- 2023 ACS Junior Technical Meeting-Puerto Rico Interdisciplinary Scientific Meeting — 29 April 2023
  University of Puerto Rico at Bayamón, Sponsored by PR-LSAMP — Bayamón, Puerto Rico
  Presentation: Properties of $\tau_{(n)}$-primes
- 38th Interuniversity Mathematical Sciences Research Seminar — 24-25 February 2023
  University of Puerto Rico, Mayagüez Campus — Mayagüez, Puerto Rico
  Presentation: Permutation Invariant Parking Assortments
- 2023 AAAS Emerging Researchers National Conference in STEM — 9-11 February 2023
  Omni Shoreham Hotel — Washington, District of Columbia
  Poster: Permutation Invariant Parking Functions with cars of assorted lengths
- Joint Mathematics Meetings 2023 — 4-7 January 2023
  John B. Hynes Veterans Memorial Convention Center — Boston, Massachusetts
  Poster: Permutation Invariant Parking Functions with cars of assorted lengths
  Presentation: Permutation Invariant Parking Functions with Cars of Arbitrary Lengths
- Field of Dreams Conference 2022 — 4-6 November 2022
  The Graduate Hotel, University of Minnesota-Twin Cities — Minneapolis, Minnesota
- 2022 SACNAS National Diversity in STEM Conference — 27-29 October 2022
  Pedro Roselló Convention Center — San Juan, Puerto Rico
  Poster: The Study of $\tau_{(n)}$-primes
- 2022 Gulf Coast Undergraduate Research Symposium — 8-9 October 2022
  William Marsh Rice University — Houston, Texas
  Presentation: Properties of $\tau_{(n)}$-primes
- Summer@ICERM 2022: Computational Combinatorics — 3 August 2022
  Institute for Computational and Experimental Research in Mathematics — Providence, Rhode Island
  Presentation: On Permutation-Invariant Parking Sequences
- 2022 ACS Junior Technical Meeting-Puerto Rico Interdisciplinary Scientific Meeting — 9 April 2022
  University of Puerto Rico at Humacao — Humacao, Puerto Rico
  Presentation: The Study of $\tau_{(n)}$-primes
- Joint Mathematics Meetings 2022 — 6-9 April 2022
  Poster: Improving Bounds of Hermitian-Lifted Codes — Virtual Conference
- 37th Interuniversity Mathematical Sciences Research Seminar — 25-26 February 2022
  Poster: The Study of $\tau_{(n)}$-primes — Virtual Conference
  Presentation: Improving Bounds of Hermitian-Lifted Codes
- 2021 Math REU Conference@Clemson University — 19 July 2021
  Presentation: Improved Hermitian-Lifted Codes — Virtual Conference

- 2021 ACS Junior Technical Meeting-Puerto Rico Interdisciplinary Scientific Meeting — 23-24 April 2021
  Presentation: The Study of $\tau_{(n)}$-atoms — Virtual Conference
- 35th Interuniversity Mathematical Sciences Research Seminar — 6-7 March 2020
  University of Puerto Rico at Cayey — Cayey, Puerto Rico
  Poster: The Study of $\tau_{(n)}$-atoms

Academics and Coursework

Purdue University

Coursework (Graduate):

MA 61500: Numerical Methods for Partial Differential Equations — January 2026–May 2026
MA 53200: Elements of Stochastic Processes — January 2026–May 2026
MA 59800ZAT: Hamiltonian Dynamics — January 2026–May 2026
MA 55400: Linear Algebra — August 2025–December 2025
MA 51900: Introduction to Probability — August 2025–December 2025
MA 57300: Numerical Solutions of Ordinary Differential Equations — August 2025–December 2025
MA 59500OT: Computational Optimal Transport and Deep Generative Models — January 2025–May 2025
MA 59800ZY: Topics in Dynamical Systems (Bifurcation Theory) — January 2025–May 2025
MA 54600: Introduction to Functional Analysis — January 2025–May 2025
MA 59500AFF: Analytic Theory of Function Fields — August 2024–December 2024
MA 59500MM: Introduction to Mathematical Modeling — August 2024–December 2024
MA 54300: Introduction to Ordinary Differential Equations and Dynamical Systems — January 2024–May 2024
MA 54400: Real Analysis and Measure Theory — January 2024–May 2024
MA 55300: Introduction to Abstract Algebra — August 2023–December 2023
MA 53000: Functions of a Complex Variable I — August 2023–December 2023

University of Puerto Rico, Mayagüez Campus

Coursework (Graduate):

MATE 6101: Number Theory I — August 2022–December 2022
MATE 5150: Linear Algebra — January 2021–May 2021

Coursework (Undergraduate):

MATE 4000: Elements of Topology — August 2022–December 2022
MATE 4052: Advanced Calculus II — January 2022–May 2022
MATE 4051: Advanced Calculus I — August 2021–December 2021
MATE 4061: Numerical Analysis I — August 2021–December 2021
MATE 4021: Introduction to Mathematical Logic — August 2021–December 2021
MATE 4010: Introduction to Complex Variables — January 2021–May 2021
ESMA 4001: Mathematical Statistics I — January 2021–May 2021
MATE 4020: Partial Differential Equations and Fourier Series — August 2020–December 2020
MATE 4008: Introduction to Algebraic Structures — August 2020–December 2020
MATE 3040: Theory of Numbers — August 2020–December 2020
MATE 4009: Ordinary Differential Equations — June 2020–July 2020
MATE 4031: Introduction to Linear Algebra — January 2020–May 2020
COMP 3010: Introduction to Computer Programming I — January 2020–May 2020
MATE 3063: Calculus III — January 2020–May 2020
MATE 3032: Calculus II — August 2019–December 2019
MATE 3020: Introduction to Mathematical Foundations — August 2019–December 2019
MATE 3031: Calculus I — June 2019–July 2019
MATE 3172: Precalculus II — January 2019–May 2019
MATE 3171: Precalculus I — August 2018–December 2018